DELTA Energy Systems

SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS)

Project	Solar Inverter monitoring with WEB'log data logger
Author	Reinhard Enste
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Revision History			
Date	Version	Description	Author(s)
14/05/2008	initial draft	Initial draft of the requirements	Reinhard Enste
16/09/2008	0.2	Add protocol CI100KW	Th. Ohnemus
30/09/2008	0.3	Backwards compatibility for old Inverters	E. Campion
08/10/2008	0.4	Complete backwards compatibility for old Inverters	Reinhard Enste
04/05/2009	0.5	Add protocol CI100KW	Th. Ohnemus
09/06/2009	0.6	Modifications to the CI100KW	E: Campion
25/06/2009	0.7	Modifications to the CI100KW Setup	E: Campion
22/10/2009	0.8	Updated some CI100KW Setup factors	E. Campion
24/11/2009	0.9	Modified CI100 Modules Error Bytes, external error bit 1, renamed in "Phase not responding" and Status byte 2 System Control, added some bits	E. Campion
01.12.2009	0.10	Add new Inverter SIxx00	Th. Ohnemus
18.01.2010	0.11	modified earth fault monitor(before Isolation and GND monitor configuration)	A. Riplinger
08.03.2010	0.12	Add new Inverter 15k/20k 3Phase TL	Th. Ohnemus
18.05.2010	0.13	Modify Inverter 15k/20k 3Phase TL	Th. Ohnemus
07.07.2010	0.14	CI100: Added per module status byte	E. Campion

24.08.2010	0.15	SOLIVIA15/20 EU TL: Supplied AC Energy Today and Total	S. Walter
26.08.2010	0.16	SOLIVIA 15 EU TL/ SOLIVIA 20 EU TL Status	S. Walter
14.09.2010	0.17	Section 3.1.4.6, added software revision description	M. Diforte
06.10.2010	0.18	Section 3.1.4.3 SI5000 measurement and statistic, correct number of Bytes	Th. Ohnemus
15.11.2010	0.19	Modifiy chapter 3.1.4.6 and 3.1.5.6 SOLIVIA EU/US/AP	Th. Ohnemus
08.12.2010	0.20	- Introduced Soliva CS and renamed CI100KW in Soliva CM/CS - Corrected error in the CM/CS measurements command: the isolation resistance value is in MOhm and has a factor 100 - CM/CS: Modified definition of the "module status byte" @ page 27	E. Campion
24.02.2011	0.21	Modified 3.4.1.4: introduced in Status System(2) bit (Cabinet OFF) that indicates if the inverter is disconnected because of somebody using the OFF button in the GUI of the SC	E. Campion
05.05.2011	0.22	Introduced 2 new error bits in the external error byte of a rack (see 3.4.1.4)	E. Campion
24.11.2011	0.23	Add the EU G4 Modells	Th. Ohnemus
08.02.2012	0.24	Correct table in chapter 3.1.4.7	Th. Ohnemus
09.02.2012	0.25	The temperatures values are always signed int16. Corrected in SI – Solivia EU/US/AP – Solivia CM/CS – Solivia 15/20 EU TL	E. Campion
14.02.2012	0.26	- Added commands "Supplied ac energy day" and "Inverter runtime day" for EU G4 Models - corrected ambient and heatsink temperature scaling to 1 = 1degree	S. Marsden
18.04.2012	0.27	Correct the chapter 3.1.4.3	Th. Ohnemus
25.04.2012	0.28	Modified 3.4.1.4: introduced in Status System(3) 2 new bits indicating possible error situations at start-up	E. Campion
12.10.2012	0.29	Add SOLIVIA 3.0 / 5.0 T4 TL Add SOLIVIA 10 / 30 T4 TL Add SOLIVIA 2.0 / 10 G4 TR Add SOLIVIA 5.0 / 12 G4 TL	S. Walter
15.11.2012	0.30	Add SOLIVIA11EUG4TR EVR	Th. Ohnemus
03.09.2013	0.31	Add SOLIVIA 6 / 8 / 12 T4 TL	S. Walter
09.09.2013	0.32	Add SOLIVIA 3.0 NAG4TL Add SOLIVIA 3.8 NAG4TL Add SOLIVIA 5.0 NAG4TL Add SOLIVIA 7.6 NAG4TL	D. Löffler
18.09.2013	0.33	Add a chapter about broadcast commands	S. Walter

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1 Introduction

1.1 Purpose

Purpose of this document is to provide the SRS for the communication interface between the Delta Solar Inverter family and the meteo control WEB'log data logger family.

This SRS will serve as
- a guideline for the developers implementing the communication protocol

1.2 Scope

We have a family of solar inverter products (SI1900/SI2500/SI3300/SI5000/...) with different hardware and software characteristics; but each product supports a standard communication protocol, the "Delta protocol", using the RS485 as physical layer.
Through the Delta protocol is possible to:
- configure the inverter (i.e. each device can have different configurations depending on the country of destination)
- activate/deactivate some functionality
- monitor the status of the inverter
- read and write statistical/measurement /production/calibration data
- update the firmware of each device (boot-loader function)

The WEB'log protocol implementation should be our solution for a remote monitor feature of all Delta solar inverter products.

1.3 Definitions, Acronyms and Abbreviations

WL: WEB'log monitor unit
SI: Solar Inverter
PV: Photovoltaic

1.4 References

[1] Solar Inverter Communication Protocol v. 3.05 by Reinhard Enste
[2] Alarm Strategy(0.5) (excel)

2 General Description

2.1 Product Perspective

This product will be a replacement of the existing communication interface between WL and the different SI devices. The new functionalities requested from marketing and solar support team to have a more useful product with the remote access to all connected SI devices.

The program will run on a WL operating system, connected to the SI devices via RS485 bus. The program will communicate with the different inverters on the bus using the Delta protocol as specified by the document [1]: the protocol itself it's a master/slave protocol where the master is the WL and the slaves are the inverters on the bus, each identified by its unique address.

Each different operation that can be performed on an addressed inverter is identified, at protocol level, by a pair command (CMD) + sub-command (SUB-CMD). The list of used CMD/SUB-CMD supported by each inverter, along with their meaning and how the transported information is codified, is specified later on in this document. Within each family (*VARIANT*), inverters with different software version and/or with different country of destination (*TYPE*) can support different command lists.

2.2 Product Functions

Once connected on the bus the WL will carry out a scan of a given address range in order to find the connected inverters along with their address and type. Each inverter use the same command set for the communication to WL. The number of data in the protocol is specific to the inverter type.

2.3 User Characteristics

These are the classes of users of the WL, reported in descending privilege levels:

- Standard (Customer)
    - User have access to a specific number of data
- Extended without write access (Customer/Service with extended access privileges)
    - User have access to a specific number of data and to more detailed service information
- Extended with write access (Customer/Service with extended access privileges)
    - User have access to a specific number of data, to more detailed service information and can change some inverter specific settings

3 Specific Requirements

3.1 Functional Requirements

The WL communicates with the inverters by using 4 different commands. The combination of CMD + SUB-CMD is always the same, but the received data length from the inverter can vary depending on the inverter variant. Only the inverter variant should used to identify the length and position of sub data from the received data packet. The next subsections give a detailed command description.

3.1.1 Inverter identification

To identify the connected inverter type the WL use the CMD [0] + SUB-CMD [0]. The data received from the inverter have the following format:

INVERTER_TYPE + INVERTER_VARIANT + "Text" + CS
Length: 1 Byte + 1 Byte + n Bytes + 1 Byte
Example:

#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
Data	6	1	S	I		2	5	0	0		D	E	,
Hex	06	01	53	49	20	32	35	30	30	20	44	45	2C

The length of the "Text" is variable depending on inverter type. To handle this, the comma separator should used to detect the end of text.

The next tables show all inverter types and variants existing at DES. The WL should handle a limited number of different inverter variants. The details about the variants described later in the document.

INVERTER_TYPE

INVERTER_TYPE	Description	INVERTER_TYPE_TEXT
0	Production release	SI xxxx PROD
1	US version	SI xxxx US
2	EU version (old VDE0126 etc.)	SI xxxx EU
3	ES Spain version	SI xxxx ES
4	IT Italian version	SI xxxx IT
5	GR Greece version	SI xxxx GR
6	DE German version (new VDE0126)	SI xxxx DE
7	PT Portugal version	SI xxxx PT
8	CZ Czech republich version	SI xxxx CZ
9	TW Taiwan	SI xxxx TW
10	BE Belgium version	SI xxxx BE

Hint!
xxxx Identifies the inverter output power i.e. 3300 = 3300 W

INVERTER_VARIANT

INVERTER_VARIANT	Description
0	Not used
1	2500 W output power
2	1800 W output power
3	3300 W output power
4	5000 W output power
5	1300 W output power
6	1670 W output power
7	1900 W output power
8	2200 W output power
99	Solivia CS 44-55-66 KW 3 ph cabinet
100	Solivia CM 77-88-100 KW 3ph cabinet

For the complete identification the WL uses the CMD[0] + SUB-CMD[64] to obtain the software version of the AC controller. How this information is codified is inverter-specific, depends in general on the variant and type previously described. At the moment we have the following situation:

- for the 3300 W (variant=3) and every possible type, the inverter responds with a frame containing 2 bytes of data: the first one is the software version minor and the second one the software version major.
  Example: 0x01 0x05 => sw version 5.1

- for the Solivia CM/CS (variant=100 or 99) the inverter responds to the sw version request with 3 bytes: the first one is the major, the second one is the minor and the third one is the bug-fix.
  Example: 0x01 0x05 0x01=> sw version 1.5.1

- for all other inverter variants and corresponding types, the inverter sends back as data always 2 bytes where the first one is the software version major and the second one the software version minor.
  Example: 0x01 0x05 => sw version 1.5

- for future variant/types the number of bytes of data and how they are codified can change.

3.1.2 Backwards compatibility

The new firmware of the WL has to be backwards compatible with old devices, which have not implemented the commands defined in this document.
The WL has to identify such devices and for them use the "old" commands. This applies to:
- SI2500 old version (variant =1 and type =2)
- SI3300 (variant=3) with sw version less than 8.0

3.1.3 Broadcast Commands for all SOLIVIA String Inverters

The SOLIVIA String Inverters are able to receive the following broadcast commands.

CMD	WSCMD	TYPE	DESCRIPTION	HINT
0	32+158	4 uc bytes	actual date	Byte 1 ->day of the week 1 .. 7 (Sunday = 1) Byte 2 -> day 1 .. 31 Byte 3 -> month 1 .. 12 Byte 4 -> year 2 .. 99
0	33+128	3 uc bytes	actual time	Byte 1 -> hours 0 .. 23 Byte 2 -> minutes 0 .. 59 Byte 3 -> seconds 0 .. 59
13	8+128	UINT8	external power control limit	1 = 1% (copy of main command 5 sub command 26)
5	31+128	INT8	Reactive power control	1 = 0.01 (negative values = capacitive operation)
15	31+128	INT16	Reactive power control	1 = 0.01 (negative values = capacitive operation)

To send a command as broadcast to all connected inverters, the destination address needs to be 0xFF (255).

3.1.4 (Solivia CM/CS) Production Data(0.6)

Only for the Solivia CM/CS we get the production data of the components of the rack with the CMD [96] and SUB-CMD[0]. Here the data received from the Solivia CM/CS:

Description	***Unit***	***# of Bytes***	***Scaling***
Rack SAP serial number		*18 Byte*	
System Controller Serial Number		*8 Bytes*	
Software revision System Controller	*MA,MI,BF*	*3 Byte*	
Hw revision System Controller		*1 Byte*	
Back Plane Controller SAP serial number		*18 Byte*	
Software revision Backplane Controller	*MA,MI,BF*	*3 Byte*	
Hardware revision Backplane Controller		*1 Byte*	
Module1 SAP serial number		*18 Byte*	
Module1 AC Controller sw version	*MA,MI,BF*	*3 Byte*	
Module1 Hardware revision		*1 Byte*	
The fields in red repeated 8 more times (in total 9 modules)		***198***	
Total # of Bytes:		***250***	

3.1.5 Measurement and statistic data

To get the measurement and statistic data from the connected inverter type the WL use the CMD [96] + SUB-CMD [1].

3.1.5.1 SI2500 measurement and statistic

Data received from SI2500 (INVERTER_VARIANT 1 and INVERTER_TYPE not 2) inverter (0.5):

Description	Unit	# of Bytes	Scaling
SAP part number		11 Byte	
SAP serial number		18 Byte	
SAP date code		4 Byte	
SAP revision		2 Byte	
Software revision ac control	MA,MI	2 Byte	
Software revision dc control	MA,MI	2 Byte	
Software revision display	MA,MI	2 Byte	
Software revision ens control	MA,MI	2 Byte	
Solar current at input 1	A	2 (MSB, LSB)	Value / 10
Solar voltage at input 1	V	2 (MSB, LSB)	
Solar isolation resistance at input 1	KOhm	2 (MSB, LSB)	
AC current	A	2 (MSB, LSB)	Value / 10
AC voltage	V	2 (MSB, LSB)	
AC power	W	2 (MSB, LSB)	
AC frequency	Hz	2 (MSB, LSB)	Value / 100
Supplied ac energy	Wh	2 (MSB, LSB)	
Inverter runtime	Minutes	2 (MSB, LSB)	
Calculated temperature at ntc (DC side)	°C	2 (MSB, LSB)	***Signed** int16*
Solar input MOV resistance	KOhm	2 (MSB, LSB)	
Calculated temperature at ntc (AC side)	°C	2 (MSB, LSB)	***Signed** int16*
AC voltage (AC control)	V	2 (MSB, LSB)	Value / 10
AC frequency (AC control)	Hz	2 (MSB, LSB)	Value / 100
DC injection current (AC control)	A	2 (MSB, LSB)	Value / 1000
AC voltage (ENS control)	V	2 (MSB, LSB)	Value / 10
AC frequency (ENS control)	Hz	2 (MSB, LSB)	Value / 100
DC injection current (ENS control)	A	2 (MSB, LSB)	Value / 1000
Maximum solar 1 input current	A	2 (MSB, LSB)	Value / 10
Maximum solar 1 input voltage	V	2 (MSB, LSB)	
Maximum solar 1 input power	W	2 (MSB, LSB)	
Minimum isolation resistance solar 1	KOhm	2 (MSB, LSB)	
Maximum isolation resistance solar 1	KOhm	2 (MSB, LSB)	
Maximum ac current of today	A	2 (MSB, LSB)	Value / 10
Minimum ac voltage of today	V	2 (MSB, LSB)	
Maximum ac voltage of today	V	2 (MSB, LSB)	
Maximum ac power of today	W	2 (MSB, LSB)	
Minimum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Maximum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Supplied ac energy	KWh	4 (MSB, LSB)	Value / 10
Inverter runtime	Hours	4 (MSB, LSB)	
Global alarm status		1 Byte	
Status DC input		1 Byte	
Limits DC input		1 Byte	
Status AC output		1 Byte	
Limits AC output		1 Byte	
Isolation warning status		1 Byte	
DC hardware failure		1 Byte	
AC hardware failure		1 Byte	
ENS hardware failure		1 Byte	
Internal Bulk failure		1 Byte	
Internal communications failure		1 Byte	
AC hardware disturbance		1 Byte	
History status messages		20 Byte	

Total # of Bytes:		**141**	

3.1.5.2 SI3300 measurement and statistic

Data received from SI3300 (INVERTER_VARIANT 3 and sw version equal or greater than 8.0) inverter (0.5):

Description	**Unit**	**# of Bytes**	**Scaling**
SAP part number		11 Byte	
SAP serial number		18 Byte	
SAP date code		4 Byte	
SAP revision		2 Byte	
Software revision ac control	MA,MI	2 Byte	
Software revision dc control	MA,MI	2 Byte	
Software revision display	MA,MI	2 Byte	
Software revision ens control master	MA,MI	2 Byte	
Software revision ens control slave	MA,MI	2 Byte	
Solar voltage at input 1	V	2 (MSB, LSB)	
Solar current at input 1	A	2 (MSB, LSB)	Value / 10
Solar isolation resistance at input 1	kOhm	2 (MSB, LSB)	
Calculated temperature at ntc (DC side)	°C	2 (MSB, LSB)	***Signed** int16*
Solar input MOV resistance	kOhm	2 (MSB, LSB)	
SI3300 Ac current	A	2 (MSB, LSB)	Value / 10
SI3300 AC voltage	V	2 (MSB, LSB)	
SI3300 AC power	W	2 (MSB, LSB)	
SI3300 AC frequency	Hz	2 (MSB, LSB)	Value / 100
SI3300 Calculated temperature at ntc (AC side)	°C	2 (MSB, LSB)	***Signed** int16*
AC ctrl voltage	V	2 (MSB, LSB)	Value / 10
AC ctrl frequency	Hz	2 (MSB, LSB)	Value / 100
ENS master Grid voltage	V	2 (MSB, LSB)	Value / 100
ENS master Grid ens frequency	Hz	2 (MSB, LSB)	Value / 100
ENS master Grid DC injection current	A	2 (MSB, LSB)	Value / 100
ENS slave Grid voltage	V	2 (MSB, LSB)	Value / 100
ENS slave Grid frequency	Hz	2 (MSB, LSB)	Value / 100
ENS slave Grid DC injection current	A	2 (MSB, LSB)	Value / 100
Supplied ac energy	Wh	2 (MSB, LSB)	
Inverter runtime	Minutes	2 (MSB, LSB)	
Maximum ac current of today	A	2 (MSB, LSB)	Value / 10
Minimum ac voltage of today	V	2 (MSB, LSB)	
Maximum ac voltage of today	V	2 (MSB, LSB)	
Maximum ac power of today	W	2 (MSB, LSB)	
Minimum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Maximum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Supplied ac energy	kWh	4 (MSB, LSB)	Value / 10
Inverter runtime	Hours	4 (MSB, LSB)	
Maximum solar 1 input current	A	2 (MSB, LSB)	Value / 10
Maximum solar 1 input voltage	V	2 (MSB, LSB)	
Maximum solar 1 input power	W	2 (MSB, LSB)	
Minimum isolation resistance solar 1	kOhm	2 (MSB, LSB)	
Maximum isolation resistance solar 1	kOhm	2 (MSB, LSB)	
Alarms status		1 Byte	
Status DC input		1 Byte	
Limits DC input		1 Byte	
Status AC output		1 Byte	
Limits AC output		1 Byte	

Warnings status		1 Byte	
DC hardware failure		1 Byte	
AC hardware failure		1 Byte	
ENS hardware failure		1 Byte	
Internal Bulk failure		1 Byte	
Internal communication failure		1 Byte	
AC hardware disturbance		1 Byte	
History status messages		20 Byte	
Total # of Bytes:		**147**	

3.1.5.3 SI5000 measurement and statistic

Data received from SI5000 (INVERTER_VARIANT 4, 14, 63) inverter: (0.5; 0.12, 0.27)

Description	Unit	# of Bytes	Scaling
SAP part number		11 Byte	
SAP serial number		18 Byte	
SAP date code		4 Byte	
SAP revision		2 Byte	
Software revision ac control	MA,MI	2 Byte	
Software revision dc control	MA,MI	2 Byte	
Software revision display	MA,MI	2 Byte	
Software revision ens control	MA,MI	2 Byte	
Solar current at input 1	A	2 (MSB, LSB)	Value / 10
Solar voltage at input 1	V	2 (MSB, LSB)	
Solar isolation resistance at input 1	kOhm	2 (MSB, LSB)	
Solar current at input 2	A	2 (MSB, LSB)	Value / 10
Solar voltage at input 2	V	2 (MSB, LSB)	
Solar isolation resistance at input 2	kOhm	2 (MSB, LSB)	
AC current	A	2 (MSB, LSB)	Value / 10
AC voltage	V	2 (MSB, LSB)	
AC power	W	2 (MSB, LSB)	
AC frequency	Hz	2 (MSB, LSB)	Value / 100
Supplied ac energy	Wh	2 (MSB, LSB)	
Inverter runtime	Minutes	2 (MSB, LSB)	
Calculated temperature at ntc (DC side)	°C	2 (MSB, LSB)	***Signed** int16*
Solar input 1 MOV resistance	kOhm	2 (MSB, LSB)	
Solar input 2 MOV resistance	kOhm	2 (MSB, LSB)	
Calculated temperature at ntc (AC side)	°C	2 (MSB, LSB)	***Signed** int16*
AC voltage (AC control)	V	2 (MSB, LSB)	Value / 10
AC frequency (AC control)	Hz	2 (MSB, LSB)	Value / 100
DC injection current (AC control)	A	2 (MSB, LSB)	Value / 1000
AC voltage (ENS control)	V	2 (MSB, LSB)	Value / 10
AC frequency (ENS control)	Hz	2 (MSB, LSB)	Value / 100
DC injection current (ENS control)	A	2 (MSB, LSB)	Value / 1000
Maximum solar 1 input current	A	2 (MSB, LSB)	Value / 10
Maximum solar 1 input voltage	V	2 (MSB, LSB)	
Maximum solar 1 input power	W	2 (MSB, LSB)	
Minimum isolation resistance solar 1	kOhm	2 (MSB, LSB)	
Maximum isolation resistance solar 1	kOhm	2 (MSB, LSB)	
Maximum solar 2 input current	A	2 (MSB, LSB)	Value / 10
Maximum solar 2 input voltage	V	2 (MSB, LSB)	
Maximum solar 2 input power	W	2 (MSB, LSB)	
Minimum isolation resistance solar 2	kOhm	2 (MSB, LSB)	

Maximum isolation resistance solar 2	kOhm	2 (MSB, LSB)	
Maximum ac current of today	A	2 (MSB, LSB)	Value / 10
Minimum ac voltage of today	V	2 (MSB, LSB)	
Maximum ac voltage of today	V	2 (MSB, LSB)	
Maximum ac power of today	W	2 (MSB, LSB)	
Minimum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Maximum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Supplied ac energy	kWh	4 (MSB, LSB)	Value / 10
Inverter runtime	Hours	4 (MSB, LSB)	
Global alarm status		1 Byte	
Status DC input		1 Byte	
Limits DC input		1 Byte	
Status AC output		1 Byte	
Limits AC output		1 Byte	
Isolation warning status		1 Byte	
DC hardware failure		1 Byte	
AC hardware failure		1 Byte	
ENS hardware failure		1 Byte	
Internal Bulk failure		1 Byte	
Internal communications failure		1 Byte	
AC hardware disturbance		1 Byte	
History status messages		20 Byte	
Total # of Bytes:		**159**	

3.1.5.4 SOLIVIA 6 / 8 / 10 / 12 / 15 / 20 / 30 TL *measurement and statistic*(0.12) (0.23) (INVERTER_VARIANT 27, 28, 43, 44, 89, 90, 91, 93, 95) inverter: (0.12) (0.23) (0.29) (0.31)

Description	Unit	# of Bytes	Scaling
SAP part number		11 Byte	
SAP serial number		18 Byte	
SAP date code		4 Byte	
SAP revision		2 Byte	
DSP FW Rev(0.13)	MA,MI	2(MSB,LSB)	
DSP FW Date	MA,MI	2(MSB,LSB)	ex : 0x0A0B => V10.11
Redundant MCU FW Rev.	MA,MI	2(MSB,LSB)	
Redundant MCU FW Date	MA,MI	2(MSB,LSB)	
Display MCU FW Rev.	MA,MI	2(MSB,LSB)	
Display MCU FW Date	MA,MI	2(MSB,LSB)	
AC Voltage(Phase1)	V	2(MSB,LSB)	Value/10
AC Current(Phase1)	A	2(MSB,LSB)	Value/100
AC Power(Phase1)	W	2(MSB,LSB)	
AC Frequency(Phase1)	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase1) **(read from Redundant)**	V	2(MSB,LSB)	Value/10
AC Frequency(Phase1) **(read from Redundant)**	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase2)	V	2(MSB,LSB)	Value/10
AC Current(Phase2)	A	2(MSB,LSB)	Value/100
AC Power(Phase2)	W	2(MSB,LSB)	
AC Frequency(Phase2)	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase2) **(Read from Redundant)**	V	2(MSB,LSB)	Value/10
AC Frequency(Phase2)	Hz	2(MSB,LSB)	Value/100

(Read from Redundant)			
AC Voltage(Phase3)	V	2(MSB,LSB)	Value/10
AC Current(Phase3)	A	2(MSB,LSB)	Value/100
AC Power(Phase3)	W	2(MSB,LSB)	
AC Frequency(Phase3)	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase3) **(Read from Redundant)**	V	2(MSB,LSB)	Value/10
AC Frequency(Phase3) **(Read from Redundant)**	Hz	2(MSB,LSB)	Value/100
Solar Voltage at Input 1	V	2(MSB,LSB)	Value/10
Solar Current at Input 1	A	2(MSB,LSB)	Value/100
Solar Power at Input 1	W	2(MSB,LSB)	
Solar Voltage at Input 2	V	2(MSB,LSB)	Value/10
Solar Current at Input 2	A	2(MSB,LSB)	Value/100
Solar Power at Input 2	W	2(MSB,LSB)	
ACPower[(0.14)]	W	2 (MSB, LSB)	
(+) Bus Voltage	V	2(MSB,LSB)	Value/10
(-) Bus Voltage	V	2(MSB,LSB)	Value/10
Supplied ac energy today[(0,15)]	Wh	4(MSB,LSB)	
Inverter runtime today	second	4(MSB,LSB)	
Supplied ac energy (total) [(0.13)(0,15)]	kWh	4(MSB,LSB)	
Inverter runtime(total)	second	4(MSB,LSB)	
Calculated temperature inside rack	°C	2(MSB,LSB)	***Signed** int16*
Status AC Output 1		1 Byte	
Status AC Output 2		1 Byte	
Status AC Output 3		1 Byte	
Status AC Output 4		1 Byte	
Status DC Input 1		1 Byte	
Status DC Input 2		1 Byte	
Error Status		1 Byte	
Error Status AC 1		1 Byte	
Global Error 1[(0.13)]		1 Byte	
CPU Error		1 Byte	
Global Error 2		1 Byte	
Limits AC output 1		1 Byte	
Limits AC output 2		1 Byte	
Global Error 3		1 Byte	
Limits DC 1		1 Byte	
Limits DC 2		1 Byte	
History status messages		20 Byte	
Total # of Bytes:		***155***	

3.1.5.5 Solivia CM/CS measurement and statistic

Data received from Solivia CM/CS (INVERTER_VARIANT 100 or 99) inverter:

Description	***Unit***	***# of Bytes***	***Scaling***
Solar current at input 1	*A*	*4(MSB, LSB)*	*Value / 100*
Solar voltage at input 1	*V*	*2 (MSB, LSB)*	
Solar isolation resistance at input 1	*MOhm*	*2 (MSB, LSB)*	*Value / 100*
Solar current at input 2	*A*	*4 (MSB, LSB)*	*Value / 100*
Solar voltage at input 2	*V*	*2 (MSB, LSB)*	
Solar isolation resistance at input 2	*MOhm*	*2 (MSB, LSB)*	*Value / 100*
AC current (Phase1)	*A*	*4 (MSB, LSB)*	*Value / 100*
AC voltage (Phase1)	*V*	*2 (MSB, LSB)*	

AC frequency (Phase1)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
AC power (Phase1)	*W*	*4 (MSB, LSB)*	
AC current (Phase2)	*A*	*4(MSB, LSB)*	*Value / 100*
AC voltage (Phase2)	*V*	*2 (MSB, LSB)*	
AC frequency (Phase2)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
AC power (Phase2)	*W*	*4 (MSB, LSB)*	
AC current (Phase3)	*A*	*4 (MSB, LSB)*	*Value / 10*
AC voltage (Phase3)	*V*	*2 (MSB, LSB)*	
AC frequency (Phase3)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
AC power (Phase3)	*W*	*4 (MSB, LSB)*	
AC power (total)	*W*	*4 (MSB, LSB)*	
Supplied ac energy today	*Wh*	*4 (MSB, LSB)*	
Inverter runtime today	*Minutes*	*2 (MSB, LSB)*	
Calculated temperature inside rack	*°C*	*2 (MSB, LSB)*	***Signed** int16*
DC injection current (AC control)	*mA*	*2 (MSB, LSB)*	
Maximum solar 1 input current	*A*	*4 (MSB, LSB)*	*Value / 100*
Maximum solar 1 input voltage	*V*	*2 (MSB, LSB)*	
Maximum solar 1 input power	*W*	*4 (MSB, LSB)*	
Minimum isolation resistance solar 1	*kOhm*	*2 (MSB, LSB)*	
Maximum isolation resistance solar 1	*kOhm*	*2 (MSB, LSB)*	
Maximum solar 2 input current	*A*	*4 (MSB, LSB)*	*Value / 100*
Maximum solar 2 input voltage	*V*	*2 (MSB, LSB)*	
Maximum solar 2 input power	*W*	*4 (MSB, LSB)*	
Minimum isolation resistance solar 2	*kOhm*	*2 (MSB, LSB)*	
Maximum isolation resistance solar 2	*kOhm*	*2 (MSB, LSB)*	
Maximum ac current of today (Phase1)	*A*	*4(MSB, LSB)*	*Value / 100*
Minimum ac voltage of today (Phase1)	*V*	*2 (MSB, LSB)*	
Maximum ac voltage of today (Phase1)	*V*	*2 (MSB, LSB)*	
Maximum ac power of today (Phase1)	*W*	*4 (MSB, LSB)*	
Minimum ac frequency of today (Phase1)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
Maximum ac frequency of today (Phase1)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
Maximum ac current of today (Phase2)	*A*	*4 (MSB, LSB)*	*Value / 100*
Minimum ac voltage of today (Phase2)	*V*	*2 (MSB, LSB)*	
Maximum ac voltage of today (Phase2)	*V*	*2 (MSB, LSB)*	
Maximum ac power of today (Phase2)	*W*	*4 (MSB, LSB)*	
Minimum ac frequency of today (Phase2)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
Maximum ac frequency of today (Phase2)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
Maximum ac current of today (Phase3)	*A*	*4 (MSB, LSB)*	*Value / 100*
Minimum ac voltage of today (Phase3)	*V*	*2 (MSB, LSB)*	
Maximum ac voltage of today (Phase3)	*V*	*2 (MSB, LSB)*	
Maximum ac power of today (Phase3)	*W*	*4(MSB, LSB)*	
Minimum ac frequency of today (Phase3)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
Maximum ac frequency of today (Phase3)	*Hz*	*2 (MSB, LSB)*	*Value / 100*
Supplied ac energy (total)	*kWh*	*4 (MSB, LSB)*	*Value / 100*
Inverter runtime (total)	*Hours*	*4 (MSB, LSB)*	
Status Solivia CM/CS (1)		*1 Byte*	
Status Solivia CM/CS (2)		*1 Byte*	
Status Solivia CM/CS (3)		*1 Byte*	
Mod 1 AC Hardware Error		*1 Byte*	
Mod 1 AC Comm. Error		*1 Byte*	
Mod 1 AC Grid Error		*1 Byte*	
Mod 1 Calibration Error		*1 Byte*	
Mod 1 DC Hardware Error		*1 Byte*	

Mod 1 DC Voltage Error		*1 Byte*	
Mod 1 I/O TWI Comm. Status		*1 Byte*	
Mod 1 External Error		*1 Byte*	
+ Red highlighted for each Module x 8		*64 Byte*	
Mod 1 Status		*1 Byte*	
+ Red highlighted for each Module x 8		*8 Byte*	
History status messages		*11 Byte*	
Total # of Bytes:		***245***	

3.1.5.6 SOLIVIA EU/US/AP Version measurement and statistic(0.10;0.19)

Data received from SOLIVIA (INVERTER_VARIANT 15 .. 23; 31 .. 39; 55 .. 60) inverter (0.10; 0.12;0.19):

Description	Unit	# of Bytes	Scaling
SAP part number		11 Byte	
SAP serial number		18 Byte	
SAP date code		4 Byte	
SAP revision		2 Byte	
Software revision ac control	Major, Minor, Bug fixing	3 Byte	
Software revision dc control	Major, Minor, Bug fixing	3 Byte	
Software revision display	Major, Minor, Bug fixing	3 Byte	
Software revision sc control	Major, Minor, Bug fixing	3 Byte	
Solar voltage at input 1	V	2 (MSB, LSB)	
Solar current at input 1	A	2 (MSB, LSB)	Value / 10
Solar isolation resistance at input 1	kOhm	2 (MSB, LSB)	
Calculated temperature at ntc (DC side)	°C	2 (MSB, LSB)	***Signed*** *int16*
Solar input MOV resistance	kOhm	2 (MSB, LSB)	
Ac current	A	2 (MSB, LSB)	Value / 10
AC voltage	V	2 (MSB, LSB)	
AC power	W	2 (MSB, LSB)	
AC frequency	Hz	2 (MSB, LSB)	Value / 100
Calculated temperature at ntc (AC side)	°C	2 (MSB, LSB)	***Signed*** *int16*
SC Grid voltage	V	2 (MSB, LSB)	Value / 100
SC Grid ens frequency	Hz	2 (MSB, LSB)	Value / 100
SC Grid DC injection current	A	2 (MSB, LSB)	Value / 100
AC Grid voltage	V	2 (MSB, LSB)	Value / 100
AC Grid frequency	Hz	2 (MSB, LSB)	Value / 100
AC Grid DC injection current	A	2 (MSB, LSB)	Value / 100
Day Supplied ac energy	Wh	2 (MSB, LSB)	(Value x 10)
Inverter runtime	Minutes	2 (MSB, LSB)	
Maximum ac current of today	A	2 (MSB, LSB)	Value / 10
Minimum ac voltage of today	V	2 (MSB, LSB)	

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Maximum ac voltage of today	V	2 (MSB, LSB)	
Maximum ac power of today	W	2 (MSB, LSB)	
Minimum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Maximum ac frequency of today	Hz	2 (MSB, LSB)	Value / 100
Supplied ac energy	kWh	4 (MSB, LSB)	Value / 10
Inverter runtime	Hours	4 (MSB, LSB)	
Maximum solar 1 input current	A	2 (MSB, LSB)	Value / 10
Maximum solar 1 input voltage	V	2 (MSB, LSB)	
Maximum solar 1 input power	W	2 (MSB, LSB)	
Minimum isolation resistance solar 1	kOhm	2 (MSB, LSB)	
Maximum isolation resistance solar 1	kOhm	2 (MSB, LSB)	
Alarms status		1 Byte	
Status DC input		1 Byte	
Limits DC input		1 Byte	
Status AC output		1 Byte	
Limits AC output		1 Byte	
Warnings status		1 Byte	
DC hardware failure		1 Byte	
AC hardware failure		1 Byte	
SC hardware failure		1 Byte	
Internal Bulk failure		1 Byte	
Internal communication failure		1 Byte	
AC hardware disturbance		1 Byte	
DC HW stage error		1 Byte	
Calibration Status		1 Byte	
Neutral Error		1 Byte	
History status messages		20 Byte	
Total # of Bytes:		**148**	

3.1.5.7 Solivia EU G4 / NA G4 measurement and statistic (0.23) (0.24) (0.26) (0.29)(0.30)

Data received from SOLIVIA (INVERTER_VARIANT 53, 73, 102, 103, 105, 106, 107, 109, 110 111, 113, 114, 120, 121, 122, 123) inverter (0.23) (0.24) (0.26):

Description	Unit	# of Bytes	Scaling
SAP part number		11 Byte	
SAP serial number		18 Byte	
SAP date code		4 Byte	
SAP revision		2 Byte	
Software Revision System controller	Major, Minor, Bug fixing	3 Byte	
Software Revision Power Controller	Major, Minor, Bug fixing	3 Byte	
Software Revision ENS controller	Major, Minor, Bug fixing	3 Byte	
Software Revision Watch dog controller	Major, Minor, Bug fixing	3 Byte	
Software Revision DC controller	Major,	3 Byte	

	Minor, Bug fixing		
Software Revision DC 1	Major, Minor, Bug fixing	3 Byte	
Software Revision DC 2	Major, Minor, Bug fixing	3 Byte	
Software Revision DC 3	Major, Minor, Bug fixing	3 Byte	
Software Revision AC	Major, Minor, Bug fixing	3 Byte	
Software Revision AC1	Major, Minor, Bug fixing	3 Byte	
Software Revision AC2	Major, Minor, Bug fixing	3 Byte	
Software Revision AC3	Major, Minor, Bug fixing	3 Byte	
Software Revision reserved	Major, Minor, Bug fixing	3 Byte	
Solar power at input 1	W	2 [UINT16]	1 = 1W
Solar voltage at input 1	V	2 [UINT16]	1 = 1V
Solar current at input 1	A	2 [UINT16]	1 = 0,1A
Solar power at input 2	W	2 [UINT16]	1 = 1W
Solar voltage at input 2	V	2 [UINT16]	1 = 1V
Solar current at input 2	A	2 [UINT16]	1 = 0,1A
Solar power at input 3	W	2 [UINT16]	1 = 1W
Solar voltage at input 3	V	2 [UINT16]	1 = 1V
Solar current at input 3	A	2 [UINT16]	1 = 0,1A
AC current - L1	A	2 [UINT16]	1 = 0,1A
AC voltage - L1	V	2 [UINT16]	1= 1V
AC frequency - L1	Hz	2 [UINT16]	1= 0,01Hz
AC active power - L1	W	2 [UINT16]	1 = 1W
AC reactive power - L1	VAR	2 [INT16]	1 = 1VAR
AC current – L2	A	2 [UINT16]	1 = 0,1A
AC voltage – L2	V	2 [UINT16]	1= 1V
AC frequency – L2	Hz	2 [UINT16]	1= 0,01Hz
AC active power – L2	W	2 [UINT16]	1 = 1W
AC reactive power – L2	VAR	2 [INT16]	1 = 1VAR
AC current – L3	A	2 [UINT16]	1 = 0,1A
AC voltage – L3	V	2 [UINT16]	1= 1V
AC frequency – L3	Hz	2 [UINT16]	1= 0,01Hz
AC active power – L3	W	2 [UINT16]	1 = 1W
AC reactive power – L3	VAR	2 [INT16]	1 = 1VAR
solar isolation plus	kOhm	2 [UINT16]	1 = 1kOhm
solar isolation minus	kOhm	2 [UINT16]	1 = 1kOhm
Temperature amb	°C	2 [INT16]	1 = 1°C
Temperature heatsink	°C	2 [INT16]	1 = 1°C

Supplied ac energy total	Wh	8 [UINT64]	1 = 1Wh
Inverter runtime total	Minutes	4 [UINT32]	1 = 1 minute
Status 1		4 [UINT32]	bit description
Status 2		4 [UINT32]	bit description
Status 3		4 [UINT32]	bit description
Status 4		4 [UINT32]	bit description
Internal status 1		4 [UINT32]	reserved bit description
Internal status 2		4 [UINT32]	reserved bit description
Internal status 3		4 [UINT32]	reserved bit description
Internal status 4		4 [UINT32]	reserved bit description
Supplied ac energy day	Wh	8 [UINT64]	1 = 1Wh
Inverter runtime day	Minutes	4 [UINT32]	1 = 1 minute
reserved		67	
Total # of Bytes:		**253**	

3.1.5.8 SOLIVIA 3.0 / 5.0 T4 TL *measurement and statistic*(0.23) (INVERTER_VARIANT 85, 88) inverter: (0.29)

Description	Unit	# of Bytes	Scaling
SAP part number		11 Byte	
SAP serial number		18 Byte	
SAP date code		4 Byte	
SAP revision		2 Byte	
DSP FW Rev	MA,MI	2(MSB,LSB)	
DSP FW Date	MA,MI	2(MSB,LSB)	ex : 0x0A0B => V10.11
Redundant MCU FW Rev.	MA,MI	2(MSB,LSB)	
Redundant MCU FW Date	MA,MI	2(MSB,LSB)	
Display MCU FW Rev.	MA,MI	2(MSB,LSB)	
Display MCU FW Date	MA,MI	2(MSB,LSB)	
Display WebPage Ctrl FW Rev.	MA,MI	2(MSB,LSB)	
Display WebPage Ctrl FW Date	MA,MI	2(MSB,LSB)	
Display WiFi Ctrl FW Rev.	MA,MI	2(MSB,LSB)	
Display WiFi Ctrl FW Date	MA,MI	2(MSB,LSB)	
AC Voltage(Phase1)	V	2(MSB,LSB)	Value/10
AC Current(Phase1)	A	2(MSB,LSB)	Value/100
AC Power(Phase1)	W	2(MSB,LSB)	
AC Frequency(Phase1)	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase1) **(read from Redundant)**	V	2(MSB,LSB)	Value/10
AC Frequency(Phase1) **(read from Redundant)**	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase2)	V	2(MSB,LSB)	Value/10
AC Current(Phase2)	A	2(MSB,LSB)	Value/100
AC Power(Phase2)	W	2(MSB,LSB)	
AC Frequency(Phase2)	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase2) **(Read from Redundant)**	V	2(MSB,LSB)	Value/10
AC Frequency(Phase2) **(Read from Redundant)**	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase3)	V	2(MSB,LSB)	Value/10
AC Current(Phase3)	A	2(MSB,LSB)	Value/100
AC Power(Phase3)	W	2(MSB,LSB)	
AC Frequency(Phase3)	Hz	2(MSB,LSB)	Value/100
AC Voltage(Phase3) **(Read from Redundant)**	V	2(MSB,LSB)	Value/10
AC Frequency(Phase3) **(Read from Redundant)**	Hz	2(MSB,LSB)	Value/100
Solar Voltage at Input 1	V	2(MSB,LSB)	Value/10
Solar Current at Input 1	A	2(MSB,LSB)	Value/100
Solar Power at Input 1	W	2(MSB,LSB)	
Solar Voltage at Input 2	V	2(MSB,LSB)	Value/10
Solar Current at Input 2	A	2(MSB,LSB)	Value/100
Solar Power at Input 2	W	2(MSB,LSB)	
ACPower	W	2 (MSB, LSB)	
(+) Bus Voltage	V	2(MSB,LSB)	Value/10
(-) Bus Voltage	V	2(MSB,LSB)	Value/10
Supplied ac energy today	Wh	4(MSB,LSB)	
Inverter runtime today	second	4(MSB,LSB)	
Supplied ac energy (total)	kWh	4(MSB,LSB)	

Inverter runtime(total)	second	4(MSB,LSB)	
Calculated temperature inside rack	캜	2(MSB,LSB)	**Signed** int16
Status AC Output 1		1 Byte	
Status AC Output 2		1 Byte	
Status AC Output 3		1 Byte	
Status AC Output 4		1 Byte	
Status DC Input 1		1 Byte	
Status DC Input 2		1 Byte	
Error Status		1 Byte	
Error Status AC 1		1 Byte	
Global Error 1		1 Byte	
CPU Error		1 Byte	
Global Error 2		1 Byte	
Limits AC output 1		1 Byte	
Limits AC output 2		1 Byte	
Global Error 3		1 Byte	
Limits DC 1		1 Byte	
Limits DC 2		1 Byte	
History status messages		20 Byte	
Total # of Bytes:		***163***	

The values of the red colored measurements (phase 2 and 3 and also the 2nd DC input) will be 0.

3.1.6 Setup data

To get the setup data from the connected inverter type the WL use the CMD [96] + SUB-CMD [2]. To overwrite the setup data with changed parameters the WL use the CMD [96] + SUB-CMD [130]. The write option should be protected with extended access privileges.

3.1.6.1 SI2500 setup

Data received from SI2500 (INVERTER_VARIANT 1 and INVERTER_TYPE not 2) inverter:

Description	Unit	# of Bytes	Scaling
Grid overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid high frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid low frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid synchronization time	s	2 (MSB, LSB)	Value / 100
Grid restart time	s	2 (MSB, LSB)	Value / 100
Grid dc current injection setpoint	A	2 (MSB, LSB)	Value / 1000
DC injection monitor		1 Byte	0 -> OFF 1 -> DC injection mode without restart 2 -> DC injection mode with restart
Isolation error setpoint	kOhm	2 (MSB, LSB)	
Varistor fuse monitor setpoint	kOhm	2 (MSB, LSB)	
Isolation error setpoint at startup	kOhm	2 (MSB, LSB)	
Isolation and GND monitor configuration		1 Byte	0xA5 > Isolation warning 0x63 -> Isolation error 0x00 -> ISO/GND OFF 0x10 -> PV+ grounded

			0x08 -> PV- grounded
MOV error enable	boolean	1 Byte	0 -> Function disabled (FALSE) 1..255 -> Function enabled (TRUE)
Store setup to EEPROM		1 Byte	1 -> Store all eeprom data
	Total # of Bytes:	**28**	

3.1.6.2 SI3300 setup

Data received from SI3300 (INVERTER_VARIANT 3 and sw version equal or greater than 8.0) inverter:

Description	Unit	# of Bytes	Scaling
Grid overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid high frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid low frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid synchronization time	s	2 (MSB, LSB)	Value / 100
Grid restart time	s	2 (MSB, LSB)	Value / 100
Grid dc current injection setpoint	A	2 (MSB, LSB)	Value / 100
Isolation error setpoint in running condition	kOhm	2 (MSB, LSB)	
Varistor monitor setpoint	kOhm	2 (MSB, LSB)	
Isolation error setpoint at startup	kOhm	2 (MSB, LSB)	
Isolation and GND monitor configuration		1 Byte	0xA5 > Isolation warning 0x63 -> Isolation error 0x00 -> ISO/GND OFF 0x10 -> PV+ grounded 0x08 -> PV- grounded
Varistor warning enable	boolean	1 Byte	0 -> Function disabled (FALSE) 1..255 -> Function enabled (TRUE)
Store setup to EEPROM		1 Byte	1 -> Store all eeprom data
	Total # of Bytes:	**27**	

3.1.6.3 SI5000 setup

Data received from SI5000 (INVERTER_VARIANT 4, 14) inverter: (0.12)

Description	Unit	# of Bytes	Scaling
Grid overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid high frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid low frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid synchronization time	s	2 (MSB, LSB)	Value / 100
Grid restart time	s	2 (MSB, LSB)	Value / 100
Grid dc current injection setpoint	A	2 (MSB, LSB)	Value /1000
DC injection monitor		1 Byte	0 -> OFF 1 -> DC injection mode without restart

			2 -> DC injection mode with restart
Isolation error setpoint	kOhm	2 (MSB, LSB)	
Isolation error setpoint at startup	kOhm	2 (MSB, LSB)	
Isolation and GND monitor configuration		1 Byte	0xA5 > Isolation warning 0x63 -> Isolation error 0x00 -> ISO/GND OFF 0x10 -> PV+ grounded 0x08 -> PV- grounded
Store setup to EEPROM		1 Byte	1 -> Store all eeprom data
Total # of Bytes:		**25**	

3.1.6.4 SOLIVIA 15 EU TL/ SOLIVIA 20 EU TL (0.12) (0.23)

Data received from SOLIVIA 15 EU TL/ SOLIVIA 20 EU TL (INVERTER_VARIANT 27, 28, 43, 44) inverter: (0.12) (0.23)

Description	Unit	# of Bytes	Scaling
Grid overvoltage setpoint	V	2(MSB,LSB)	Value/10
Grid overvoltage trip time	s	2(MSB,LSB)	Value/100
Grid overvoltage recovery setpoint	V	2(MSB,LSB)	Value/10
Grid undervoltage setpoint	V	2(MSB,LSB)	Value/10
Grid undervoltage trip time	s	2(MSB,LSB)	Value/100
Grid undervoltage recovery setpoint	V	2(MSB,LSB)	Value/10
Grid overvoltage setpoint slow	V	2(MSB,LSB)	Value/10
Grid overvoltage trip time slow	s	2(MSB,LSB)	Value/100
Grid overvoltage recovery setpoint slow	V	2(MSB,LSB)	Value/10
Grid undervoltage setpoint slow	V	2(MSB,LSB)	Value/10
Grid undervoltage trip time slow	s	2(MSB,LSB)	Value/100
Grid undervoltage recovery setpoint slow	V	2(MSB,LSB)	Value/10
Grid high frequency setpoint	Hz	2(MSB,LSB)	Value/100
Grid high frequency trip time	s	2(MSB,LSB)	Value/100
Grid high frequency recovery setpoint	Hz	2(MSB,LSB)	Value/100
Grid low frequency setpoint	Hz	2(MSB,LSB)	Value/100
Grid low frequency trip time	s	2(MSB,LSB)	Value/100
Grid low frequency recovery setpoint	Hz	2(MSB,LSB)	Value/100
DC Injection current	A	2(MSB,LSB)	Value/100
DC Injection Time	s	2(MSB,LSB)	Value/10
Active islanding gain		2(MSB,LSB)	
Short Restart Time (grid fault < 3 sec)	s	2(MSB,LSB)	Value/10
Insulation R Setpoint	1K	2(MSB,LSB)	
restart time	s	2(MSB,LSB)	Value/10
Grid synchronization time	s	2(MSB,LSB)	Value/10
Ground Current	A	2(MSB,LSB)	Value/10
Ground Current Time	s	2(MSB,LSB)	Value/10
Max Power	1W	2(MSB,LSB)	
Country_Configuration_Reg		2(MSB,LSB)	0: Insulation disable 1: Insulation enable

Store setup to EEPROM		1 Byte	1 -> Store all eeprom data

3.1.6.5 Solivia CM/CS setup

Data received from Solivia CM/CS (INVERTER_VARIANT 100 or 99) inverter:
:

Description	Unit	# of Bytes	Scaling
Grid overvoltage setpoint	V	2 (MSB, LSB)	Value
Grid critical overvoltage setpoint	V	2 (MSB, LSB)	Value
Grid undervoltage setpoint	V	2 (MSB, LSB)	Value
Grid critical undervoltage setpoint	V	2 (MSB, LSB)	Value
Grid high frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid low frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid synchronization time	s	2 (MSB, LSB)	Value / 100
Grid restart time	s	2 (MSB, LSB)	Value / 100
Grid dc current injection setpoint	mA	2 (MSB, LSB)	Value
Isolation error setpoint at startup String 1	MOhm	2 (MSB, LSB)	Value / 100
Isolation and GND monitor configuration String 1		1 Byte	0x00> Isolation warning 0x01 -> Isolation error 0x02 -> ISO/GND OFF 0x03 -> PV+ grounded 0x04 -> PV- grounded
Isolation error setpoint at startup String 2	MOhm	2 (MSB, LSB)	Value / 100
Isolation and GND monitor configuration String 2		1 Byte	0x00> Isolation warning 0x01 -> Isolation error 0x02 -> ISO/GND OFF 0x03 -> PV+ grounded 0x04 -> PV- grounded
Total # of Bytes:		**23**	

3.1.6.6 SOLIVIA EU/US/AP Version setup(0.10; 0.11; 0.12;0.19)

Data received from SOLIVIA (INVERTER_VARIANT 15 .. 23; 31 .. 39; 55 .. 60) inverter (0.10; 0.11;0.12;0.19):

Description	Unit	# of Bytes	Scaling
Grid overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical overvoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid critical undervoltage setpoint	V	2 (MSB, LSB)	Value / 10
Grid high frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid low frequency setpoint	Hz	2 (MSB, LSB)	Value / 100
Grid synchronization time	s	2 (MSB, LSB)	Value / 100
Grid dc current injection setpoint	A	2 (MSB, LSB)	Value / 100
Isolation error setpoint in running condition	kOhm	2 (MSB, LSB)	

Varistor monitor setpoint	kOhm	2 (MSB, LSB)	
Isolation error setpoint at startup	kOhm	2 (MSB, LSB)	
~~Isolation and GND monitor configuration~~ Earth fault monitor		1 Byte	For EU: Value = 1 : enable Iso error, disable varistor Value = 33 : enable Iso error, enable varistor Value= 2 : enable Iso warning, disable varistor Value= 34 : enable Iso warning, enable varistor Value = 4 : PV plus grounded. disable varistor Value = 36 : PV plus grounded. enable varistor Value = 8 : PV minus grounded, disable varistor Value = 40 : PV minus grounded, enable varistor Value = 16 : disable iso and gnd, disable varistor Value = 48 : disable iso and gnd, enable varistor For US: Value = 72 : enable GFDI and fuse to plus grounded, disable varistor Value = 104 : enable GFDI and fuse to plus grounded, enable varistor Value = 68 : enable GFDI and fuse to minus grounded, disable varistor Value = 100 : enable GFDI and fuse to minus grounded, enable varistor
~~Varistor warning enable~~	~~boolean~~	~~1 Byte~~	~~0 -> Function disabled (FALSE)~~ ~~1..255 -> Function enabled (TRUE)~~
Store setup to EEPROM		1 Byte	1 -> Store all eeprom data
Total # of Bytes:		~~**25**~~ **24**	

3.1.6.7 SOLIVIA EU G4 / NA G4 Models Version setup(0.23)(0.24) (0.26)(0.30)

Data received from SOLIVIA (INVERTER_VARIANT 53, 73, 102, 103, 105, 106, 107, 109, 110, 111, 113, 114, 120, 121, 122 and 123)
inverter (0.23)(0.24) (0.26) :

Description	**Unit**	**# of Bytes**	**Scaling**

Reactive power mode		1 [UINT8]	0 = Reactive power disabled 1 = Fix power factor 2 = Power factor charateristic Cosφ(P) 3 = Fix reactive power 4 = Reactive power charateristic Q(U)
Adjustable maximal allow output power		1 [UINT8]	1 = 0.01Pn
Recover frequency level		2 [UINT16]	1 = 0.01Hz
Fix power factor		2 [INT16]	1 = 0.01
Power factor point 1 of power factor curve (PF1)		2 [INT16]	1 = 0.01
Power factor point 2 of power factor curve (PF2)		2 [INT16]	1 = 0.01
Power factor point 3 of power factor curve (PF3)		2 [INT16]	1 = 0.01
Power factor point 4 of power factor curve (PF4)		2 [INT16]	1 = 0.01
Active power point 2 of power factor curve (AP2)		1 [UINT8]	1 = 0.01Pn
Active power point 3 of power factor curve (AP3)		1 [UINT8]	1 = 0.01Pn
Fix reactive power		2 [INT16]	1 = 0.01Pn
Q(u) maximal reactive power (cap) (Q/Pn)		2 [INT16]	1 = 0.01Pn
Q(u) minimal reactive power (ind) (Q/Pn)		2 [INT16]	1 = 0.01Pn
Q(u) Voltage to maximal reactive power		2 [UINT16]	1 = 1V
Q(u) Voltage to minimal reactive power		2 [UINT16]	1 = 1V
The width of hysteresis band		2 [UINT16]	1 = 1V
Delay time		2 [UINT16]	1 = 10ms
K factor		1 [UINT8]	1 = 1
Power limit until power off where reactive power must be set to 0.		1 [UINT8]	1 = 0.01Pn
Upper voltage limit of the dead band		2 [UINT16]	1 = 1V
Under voltage limit of the dead band		2 [UINT16]	1 = 1
FRT active time after voltage comes back to dead band.		2 [UINT16]	1 = 10ms
FRT_IBLimitSymetricFail		1 [UINT8]	1 = 1%
FRT_IBLimitASymetricFail		1 [UINT8]	1 = 1%
Frequency level to start reduction active power		2 [UINT16]	1 = 0.01Hz
Frequency level to stop reduction active power		2 [UINT16]	1 = 0.01Hz
Power reduction gradient depend on frequency		1 [UINT8]	1 = 1%
reserved		208	ignore
Total # of Bytes:		**253**	

3.2 PowerControl and Control of cos(Phi) (0.5) (0.23)

Regarding the EEG-Novelle for System bigger than 100kWp installed power. The EVU can send via remote control the set point for power limitation.
ON / OFF: 100% / 0% or reduced power: 100%, 60%, 30% an 0%.

3.2.1.1 Global call / Broadcast (0.5)

Main	Sub	Symbol	Description	Unit / Format	# of Data-bytes	Scaling	Read	Write	Default data	Setting range
13	136		Maximum power in percentage	%	1 Byte			x	100	TBD..100

3.2.1.2 Global call / Broadcast Control of cos(Phi) (0.23)

Control of cos(Phi) only for SOLIVIA (INVERTER_VARIANT 43, 44, 85, 88, 91, 95) inverter (0.23)

Main	Sub	Symbol	Description	Unit / Format	# of Data-bytes	Scaling	Read	Write	Default data	Setting range
15	159		Fixed power factor cos ϕ	%	2 (MSB, LSB)	0,01		x	Fixed default values acc to user selection: 0.95 -> MVD, 0.95 -> LVD for unit 3.68 … 13.8kVA, 0.90 -> LVD for unit > 13.8kVA	0.80(cap) ~ 1.00, 0.80(ind) ~ 1.00, Max Delay = 10sec

3.2.1.3 Global call / Broadcast Control of cos(Phi) (0.23)

Control of cos(Phi) only for SOLIVIA (INVERTER_VARIANT 53, 73, 102 103, 105, 106, 107, 109, 110, 111, 113) inverter (0.23) :

Main	Sub	Symbol	Description	Unit / Format	# of Data-bytes	Scaling	Read	Write	Default data	Setting range
5	159		Fixed power factor cos ϕ	%	2 [INT16]	0,01				0.80(cap) ~ 1.00, 0.80(ind) ~ 1.00, Max Delay =

										10sec

3.2.1.4 Feedback from Inverter (0.5)

The Inverter send a feedback
In the Statusbyte:

SC HW Disturbance, Bit 1, Ext. Power Control active

Bit	SC HW Disturbance
08	RTC error
07	AC relay error
06	reserved
05	reserved
04	reserved
03	reserved
02	IT autotest fail
01	**Ext. Power Control active**

a.) If the Inverter is running in normal condition and is able to deliver more Power.
b.) The Inverter is in thermal derating (maybe 80%) and the EVU send the value reduce to 60%

This bit is only set when the Inverter had to reduce the Power although to Inverter is able to delivere more Power.

With this Information the user can identify if the solar plant had profit cut because of the EVU request.

3.3 Status description

3.3.1.1 SI2500 status

4	DC1 overvoltage	
3	reserved	
2	reserved	
1	reserved	
0	reserved	

Delta Energy Systems

Alarms (Error # 50x)	
8	DC1 startup isolation failure
7	reserved
6	DC1 operate isolation failure
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Status DC	
8	Internal DC hardware failure
7	Internal communication error
6	Internal bulk failure
5	Solar power too low
4	Temperature limited operation
3	Power limited operation
2	MPP limited operation
1	Limits DC

Limits DC (Error # 40x)	
8	DC1 too low
7	DC1 critical undervoltage
6	DC1 undervoltage
5	DC1 overvoltage
4	reserved
3	reserved
2	reserved
1	reserved

Bulk Error	
8	Undervoltage
7	Overvoltage
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Communication Error	
8	AC to DC failure
7	AC to ENS failure
6	AC to Display failure
5	reserved
4	AC to DC error
3	AC to ENS error
2	AC to Display error
1	reserved

DC Hardware Error	
8	NTC defect
7	NTC overtemperature
6	reserved
5	reserved
4	MOV failure
3	reserved
2	reserved
1	reserved

Status AC	
8	Internal AC hardware failure
7	Internal ENS hardware failure
6	Internal AC hardware disturbance
5	reserved
4	Normal operation
3	Sync to AC
2	reserved
1	Limits AC

Limits AC (Error # 20x)	
8	Critical undervoltage
7	Undervoltage
6	Overvoltage
5	Critical overvoltage
4	Low frequency
3	High frequency
2	Islanding
1	DC injection current

AC Hardware Disturbance	
8	RTC error
7	AC Relay error
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

ENS Hardware Error	
8	Hardware error
7	reserved
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

AC Hardware Error	
8	NTC defect
7	NTC overtemperature
6	reserved
5	Helpingvoltage failure
4	Overcurrent shutdown
3	Calibration data corrupted
2	reserved
1	Max. AC power shutdown

3.3.1.2 SI3300 status

Alarms Status

Bit	Description
8	Startup isolation failure
7	reserved
6	Running isolation failure
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Warnings Status

Bit	Description
8	Reserved
7	Start up input isolation status
6	PV + grounded status
5	Running input solation status
4	PV - grounded status
3	Reserved
2	Varistor status
1	Reserved

Status DC Input

Bit	Description
8	Internal DC hardware failure
7	Internal communication error
6	Internal bulk failure
5	Input power low
4	Temperature limited operation
3	Power limited operation
2	MPP limited operation
1	Limits DC

DC Hardware Failure

Bit	Description
8	NTC defect
7	NTC overtemperature
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Internal Communication Failure

Bit	Description
8	AC to DC failure
7	AC to ENS master failure
6	AC to ENS slave failure
5	AC to Display failure
4	AC to DC error
3	AC to ENS master error
2	AC to ENS slave failure
1	AC to Display failure

Internal Bulk Failure

Bit	Description
8	Undervoltage
7	Overvoltage
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Limits DC input

Bit	Description
8	Input voltage too low
7	Input critical undervoltage
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Status AC Output

Bit	Description
8	Internal AC hardware failure
7	Internal ENS hardware failure
6	Internal AC hardware disturbance
5	reserved
4	Normal operation
3	Sync to AC
2	Calibration on going
1	Limits AC

AC Hardware Failure

Bit	Description
8	NTC defect
7	NTC overtemperature
6	HW revision missmatch
5	Helpingvoltage
4	Overcurrent shutdown
3	Calibration data corrupted
2	reserved
1	reserved

ENS Hardware Failure

Bit	Description
8	Hardware error
7	reserved
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

AC HW Disturbance

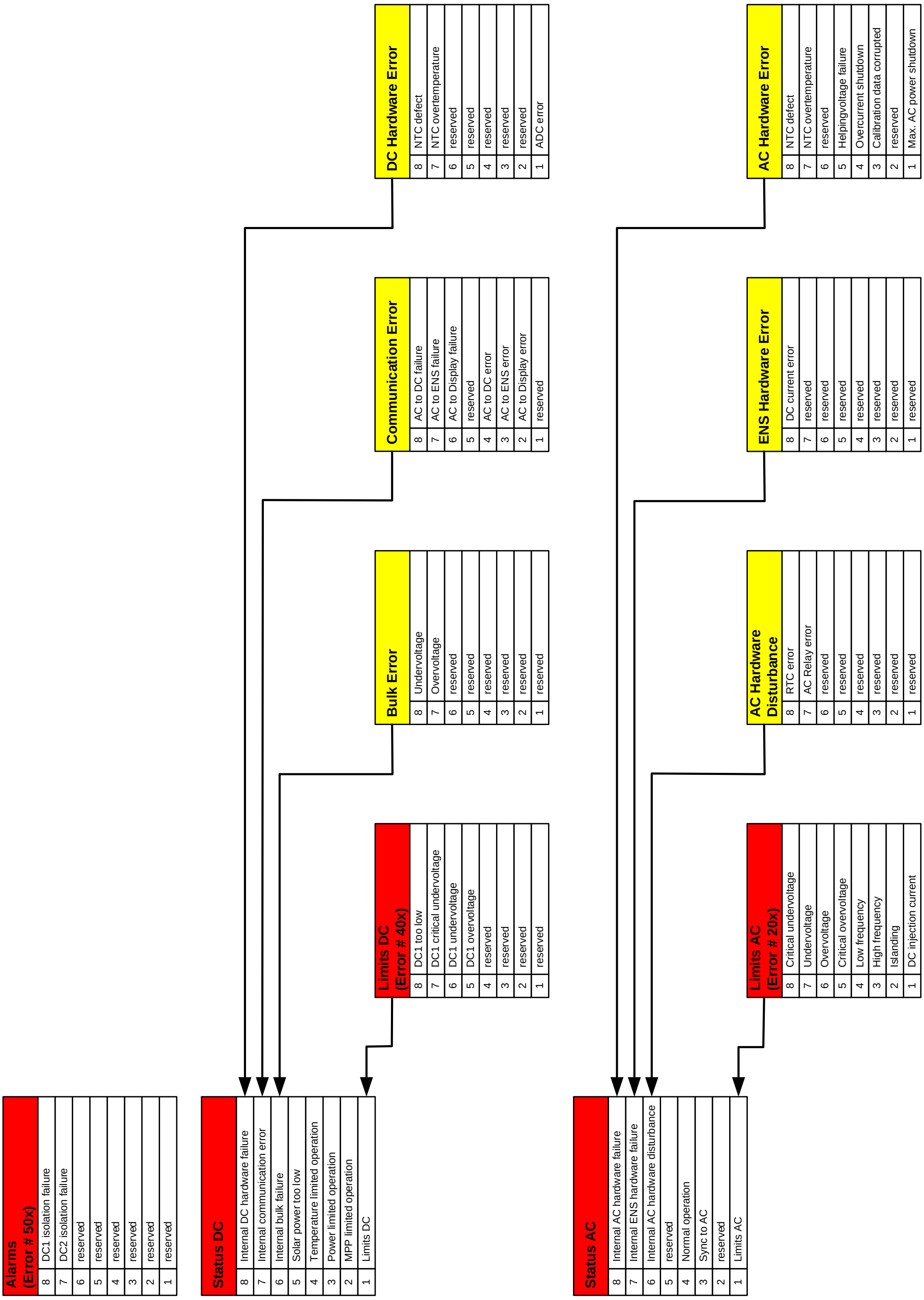
Bit	Description
8	RTC error
7	Internal relay fault
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Limits AC output

Bit	Description
8	Critical undervoltage
7	Undervoltage
6	Overvoltage
5	Critical overvoltage
4	Low frequency
3	High frequency
2	reserved
1	DC injection current

![DELTA Energy Systems]

3.3.1.3 SI5000 status



3.3.1.4 Solivia CM/CS status(0.5)

For the Solivia CM/CS we use 3 bytes for the Status of the complete System and for the 9 Modules we use 9 bytes for each. In Sum 81 bytes for Status information.

Status – System Controller

Status System (1)	
8	Isolation error String 2
7	Isolation error String 1
6	Reserved
5	Varistor DC demaged
4	Varistor AC demaged
3	Quint3 20A
2	Quint3 3.5A
1	BPC communication error

Status System (2)	
8	reserved
7	reserved
6	Cabinet OFF
5	Cabinet temperature error
4	Fan 3 blocked
3	Fan 2 blocked
2	Fan 1 blocked
1	PowerControl

Status System (3)	
8	reserved
7	reserved
6	reserved
5	reserved
4	reserved
3	reserved
2	Date automatically set at start-up
1	BPC not respondig at start-up

In Status System (2) bit 6 gets to 1 every time the cabinet is disconnected from the grid using the GUI of the System Controller. As soon as is re-connected the bit gets to 0. This happens from software version 0.3.0 of the SC. In Status System(3) bit 1 get to 1 if the BPC does not answer to the SC at start-up, and bit 2 if the SC finds data on the CF referring to a date in the future and the SC does not get any feedback from the user for 10 minutes. The bits are set at start-up and never reset. They are active from software version 0.3.3.

The 8 Error bytes for each Module.

Errors - Module

AC Hardware Errors	
7	NTC defect
6	NTC overtemperature
5	HW revision missmatch
4	Helpingvoltage
3	Overcurrent shutdown
2	Calibration data corrupted
1	Bulk under voltage
0	Bulk over voltage

DC Hardware Error	
7	NTC defect
6	NTC overtemperature
5	ENS_OK error
4	AC_OK error
3	reserved
2	Hardware error
1	Buck error
0	Bridge error

AC Comm. Errors	
7	Ext. Comm. timeout
6	IO fail
5	DC fail
4	ENS fail
3	IO broken
2	DC broken
1	ENS broken
0	reserved

DC Voltage error	
7	reserved
6	reserved
5	reserved
4	reserved
3	reserved
2	Input Over voltage
1	input voltage start up
0	Input Under voltage

AC GRID ERRORS	
7	Critical undervoltage
6	Undervoltage
5	Overvoltage
4	Critical overvoltage
3	Low frequency
2	High frequency
1	reserved
0	DC injection current

IO TWI comm status	
7	reserved
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	IO TWI bus error
0	IO Com error

Calibration errors	
7	reserved
6	reserved
5	DC HW revision error
4	AC HW revision error
3	Relay error
2	ENS compensation error
1	AC compensation error
0	LEM calibration error

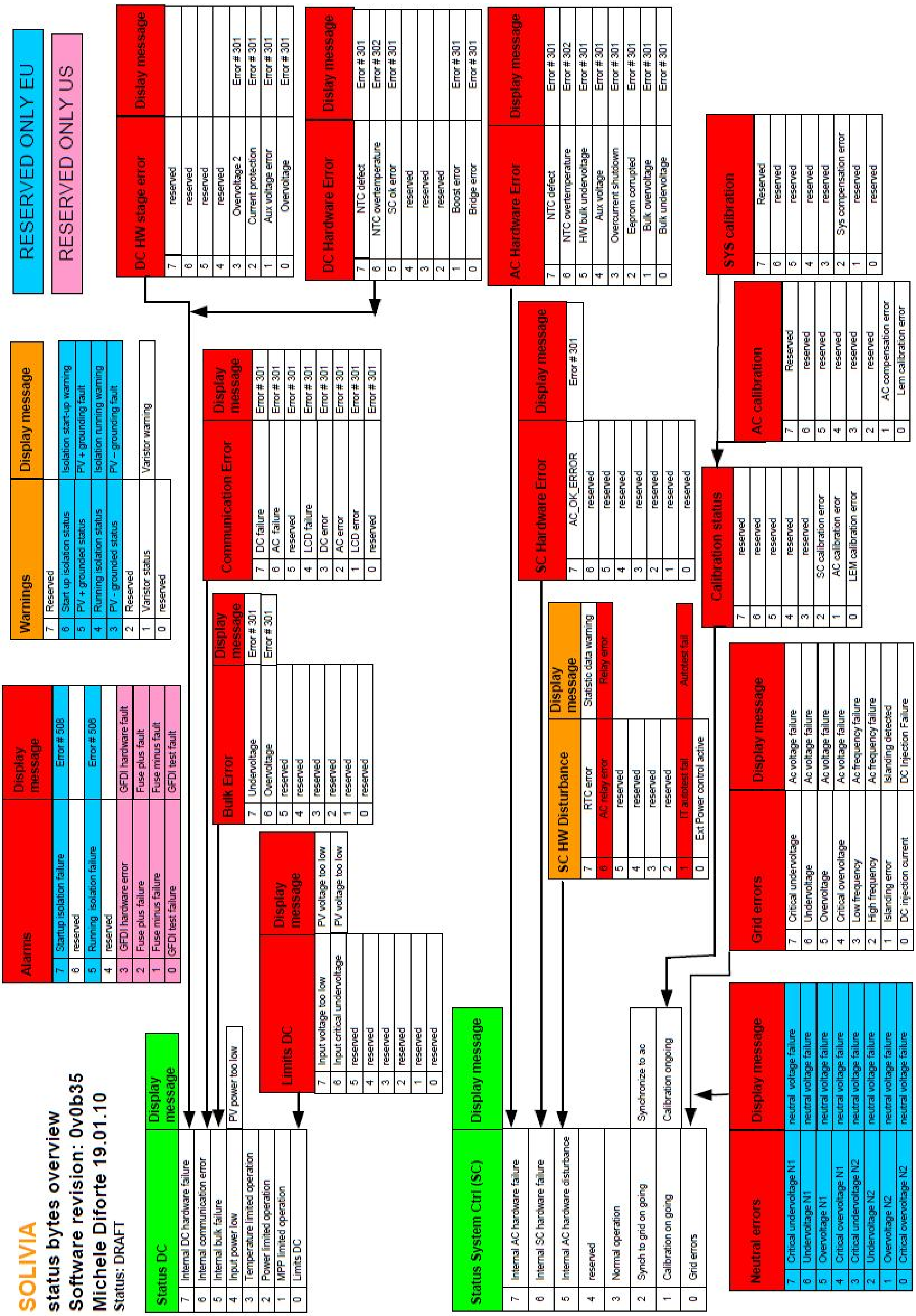
External Error	
7	reserved
6	reserved
5	reserved
4	reserved
3	Power unsymmetric
2	I/O discrepancy
1	Phase not responding
0	Modul not responding

1 Status byte for each module

Status byte	
7	Master of Bus
6	Master of MPPT
5	reserved
4	Off
3	Error
2	Working
1	Ready
0	Present/not present

0 -> not present
1 -> present

SOLIVIA Status(0.11)



3.3.1.5 SOLIVIA 3 / 5 / 6 / 8 / 10 / 12 / 15 / 20 / 30 EU TL Status(0.12) (0.16) (0.29) (0.31)

Status AC Output 1 Command: 64-1	
8	Grid Quality
7	Phase Jump
6	reserved
5	reserved
4	reserved
3	Under Frequency Range
2	Over Frequency Range
1	reserved

Status AC Output 2 Command: 64-2	
8	Under Voltage Range_S
7	reserved
6	Slow Over Voltage Range_R
5	Over Transient Voltage Range_R
4	Over Voltage Range_R
3	Under Voltage Range_R
2	No Grid
1	AC connected Fail

Status AC Output 3 Command: 64-3	
8	Slow Over Voltage Range_T
7	Over Transient Voltage Range_T
6	Over Voltage Range_T
5	Under Voltage Range_T
4	reserved
3	Slow Over Voltage Range_S
2	Over Transient Voltage Range_S
1	Over Voltage Range_S

Status AC Output 4 Command: 64-4	
8	reserved
7	reserved
6	reserved
5	reserved
4	reserved
3	reserved
2	3 Phases Unbalance
1	reserved

Status DC Input 1 Command: 64-5	
8	String2 PV input voltage too high
7	String1 PV input voltage too high
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	reserved

Status DC Input 2 Command: 64-6	
8	reserved
7	reserved
6	reserved
5	reserved
4	reserved
3	SOLAR_2_UVR
2	SOLAR_1_UVR
1	reserved

Error Status Command: 65-1	
8	reserved
7	reserved
6	reserved
5	reserved
4	FAN_FAIL
3	reserved
2	reserved
1	reserved

Error Status AC 1 Command: 65-2	
8	Heat Sink over temperature
7	Inside NTC Circuit Fail
6	Inside OTP
5	DC Injection_Total
4	DC Injection_T
3	DC Injection_S
2	DC Injection_R
1	reserved

Global Error 1 Command: 65-3	
8	DSP ADC Vgrid/Iout Bias Fail
7	Firmware Incompatibility
6	Relay Open
5	reserved
4	Inverter Choke over temperature
3	Heat Sink Ntc3 Circuit Fail
2	Heat Sink Ntc2 Circuit Fail
1	Heat Sink Ntc1 Circuit Fail

CPU Error Command: 65-4	
8	Internal Communication Fault (between Display)
7	Internal Communication Fault (between Redundant)
6	Fan Fail
5	Efficiency Abnormal
4	Red. ADC Iout_dc Bias Fail
3	Red. ADC Vgrid/Vinv Bias Fail
2	DSP ADC Iin/Iboost Bias Fail
1	DSP ADC Vin/Vbus Bias Fail

Global Error 2 Command: 65-5	
8	Bus_P over voltage rating
7	Bus unbalance
6	Relay Test Open
5	Relay Test Short
4	RCMU Circuit Fail
3	3P Current Unbalance
2	Insulation
1	Residual Current Over Rating

Limits AC output 1 Command: 66-1	
8	Output Current Over Rating_S
7	Output Current Transient Over Rating_S
6	Output Current Over Rating_R
5	Output Current Transient Over Rating_R
4	Bus voltage over rating
3	Bus_N under voltage rating
2	Bus_N over voltage rating
1	Bus_P under voltage rating

Limits AC output 2 Command: 66-2	
8	reserved
7	Inverter Failure
6	HW OOCP circuit
5	CT current sensor Fail_C
4	CT current sensor Fail_B
3	CT current sensor Fail_A
2	Output Current Over Rating_T
1	Output Current Transient Over Rating_T

Global Error 3 Command: 66-3	
8	reserved
7	reserved
6	reserved
5	reserved
4	reserved
3	Zero Cross Circuit Fail
2	reserved
1	reserved

Limits DC 1 Command: 66-4	
8	reserved
7	reserved
6	PV2 Current Over Rating
5	PV1 Current Over Rating
4	reserved
3	reserved
2	reserved
1	reserved

Limits DC 2 Command: 66-5	
8	PV2 Current Transient Over Rating
7	PV1 Current Transient Over Rating
6	reserved
5	reserved
4	reserved
3	reserved
2	reserved
1	power control active

3.3.1.6 SOLIVIA 11 EU G4 Models Version – Bit Description (0.23)

Status 1

bit 0 = 1 -> Self test ongoing
bit 1 = 1 -> Firmware update
bit 2 = 1 -> Night mode
bit 3 = 1 -> L1 Voltage failure
bit 4 = 1 -> L2 Voltage failure
bit 5 = 1 -> L3 Voltage failure
bit 6 = 1 -> L1 Frequency failure
bit 7 = 1 -> L2 Frequency failure
bit 8 = 1 -> L3 Frequency failure
bit 9 = 1 -> L1 DC Inj. failure
bit 10 = 1 -> L2 DC Inj. Failure
bit 11 = 1 -> L3 DC Inj. failure
bit 12 = 1 -> L1 islanding
bit 13 = 1 -> L2 islanding
bit 14 = 1 -> L3 islanding
bit 15 = 1 -> L1 Grid error
bit 16 = 1 -> L2 Grid error
bit 17 = 1 -> L3 Grid error
bit 18 = 1 -> L1 Long grid out
bit 19 = 1 -> L2 Long grid out
bit 20 = 1 -> L3 Long grid out
bit 21 = 1 -> L1 Grid synchronization error
bit 22 = 1 -> L2 Grid synchronization error
bit 23 = 1 -> L3 Grid synchronization error
bit 24..31 RESERVED

Status 2

bit 0 = 1 -> PV1 Iso startup failure
bit 1 = 1 -> PV1 Iso running failure
bit 2 = 1 -> PV1+ grounding failure
bit 3 = 1 -> PV1- grounding failure
bit 4 = 1 -> PV2 Iso startup failure
bit 5 = 1 -> PV2 Iso running failure
bit 6 = 1 -> PV2+ grounding failure
bit 7 = 1 -> PV2- grounding failure
bit 8 = 1 -> PV3 Iso startup failure
bit 9 = 1 -> PV3 Iso running failure
bit 10 = 1 -> PV3+ grounding failure
bit 11 = 1 -> PV3- grounding failure
bit 12 = 1 -> PV1 voltage too low failure
bit 13 = 1 -> PV2 voltage too low failure
bit 14 = 1 -> PV3 voltage too low failure
bit 15 = 1 -> Internal failure
bit 16 = 1 -> Auto test failure
bit 17 = 1 -> PV power too low failure
bit 18..31 RESERVED

Status 3

bit 0 = 1 -> PV1 Iso startup warning
bit 1 = 1 -> PV1 Iso running warning
bit 2 = 1 -> PV1+ grounding warning
bit 3 = 1 -> PV1- grounding warning
bit 4 = 1 -> PV2 Iso startup warning
bit 5 = 1 -> PV2 Iso running warning
bit 6 = 1 -> PV2+ grounding warning
bit 7 = 1 -> PV2- grounding warning
bit 8 = 1 -> PV3 Iso startup warning
bit 9 = 1 -> PV3 Iso running warning
bit 10 = 1 -> PV3 + grounding warning
bit 11 = 1 -> PV3- grounding warning
bit 12 = 1 -> PV1 voltage too low warning
bit 13 = 1 -> PV2 voltage too low warning
bit 14 = 1 -> PV3 voltage too low warning
bit 15..31 RESERVED

Status 4

bit 0 = 1 -> Internal fan warning
bit 1 = 1 -> External fan warning
bit 2 = 1 -> Synchronization ongoing
bit 3..31 RESERVED

3.4 History Status Description

The history status is decoded in the format:

```
DC hardware failure               0x10bb       bb = DC Hardware Error Bits
Internal communication failure    0x20bb       bb = Communication Error Bits
DC limitations                    0x30bb       bb = Dc Limits Bits
AC hardware failure               0x40bb       bb = AC Hardware Error Bits
ENS hardware failure              0x50bb       bb = ENS Hardware Error Bits
Grid failure                      0x60bb       bb = Ac Limits Bits
Alarm status                      0x70bb       bb = Alarm (Error #50x) Bits
Bulk error                        0x80bb       bb = Bulk Error Bits
```

Details about the different bits positions are described in the chapter *status description*.

3.5 Concerns / Queries / Doubts (if any)

[You can put in your question and doubts. These queries will be answered and will be put up in the discussion.]